Journal of Environmental Management 335 (2023) 117579



Contents lists available at ScienceDirect

Journal of Environmental Management

journal homepage: www.elsevier.com/locate/jenvman



Research article



Optimal sensor placement for the routine monitoring of urban drainage systems: A re-clustering method

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ARTICLE INFO

Keywords:
Urban drainage system
Sensor placement
Clustering analysis
Information theory

ABSTRACT

The construction of an efficient monitoring network is critical for the effective and safe management of urban drainage systems. This study developed a re-clustering methodology that incorporates additional perspectives beyond node similarity to improve the traditional clustering process for optimal sensor placement. Instead of targeting event-specific water quality or hydraulic monitoring, the method integrates the water hydraulic and quality characteristics of nodes in response to the demand for routine monitoring. The implementation of this method first applies model simulation to generate the attribute datasets required for clustering analysis, and then re-clusters the initial clustering result according to the constructed re-clustering potential indices. And the information theory-based evaluation metrics were introduced to quantitatively assess the sensor deployment scheme obtained by amalgamating the two clustering results. Two networks with different drainage systems and sizes were chosen as case studies to illustrate the application of the framework. The results demonstrate that the clustering process enables to expand the information contained in the monitoring network, and that the re-clustering strategy can generate more comprehensive and practical solutions upon this basis.

1. Introduction

An Urban Drainage System (UDS) is a critical infrastructure that collects urban runoff in catchments, conveys stormwater to receiving water bodies and sewer effluent to wastewater treatment plants, alleviates inundation severity on urban surfaces and regulates urban water pollution. The implementation of a monitoring network is crucial for the efficient and safe operation of UDS (Casal-Campos et al., 2018). Specifically in the following three aspects: firstly, water level and flow monitoring for drainage systems provide timely information on the drainage capacity and overflow risk, significantly reducing potential economic losses and threats to public life from urban flooding (Eui et al., 2016; Li et al., 2020; Li, 2021); secondly, the routine monitoring of drainage networks accumulates long-term dynamic operating conditions, and the monitoring data can be processed and analyzed to identify and diagnose abnormal operating conditions including corrosion and damage, misconnection and mixed connection of pipelines, which may result in water pollution and road collapse (Almeida and Brito, 2002; Miao et al., 2018; Xu et al., 2014); thirdly, water quality monitoring increases the probability of early detection of water quality deterioration and pollution source identification (Banik et al., 2015; Shao et al., 2021; Yang and Boccelli, 2016).

Aside from the upgrading of monitoring equipment and technology, the positioning of monitoring stations is essential to the construction of an efficient monitoring network. Since the number of sensors is generally limited by constraints such as budget, installation and maintenance, an effective monitoring network has to be adequately designed by optimizing the position of sensors (Didrik et al., 2018). Due to a variety of operational challenges with the UDS's monitoring network (Thompson et al., 2011)—e.g. complex sediment behavior, clogging or sensor mechanical damage by debris, sensor fouling, and fluctuating water levels, investigations on the optimal sensor placement emerged relatively late. With the development of monitoring devices and online monitoring technologies, the necessity to optimize the sensor arrangement in UDS has become increasingly evident in recent years.

When the sensor placement problem is solved to detect illicit polluting discharges in UDS, single-objective or multi-objective optimization models are generally formulated based on the information

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https://doi.org/10.1016/j.jenvman.2023.117579
Received 7 December 2022; Received in revised form 4 February 2023; Accepted 22 February 2023
Available online 26 February 2023
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theory. Banik et al. (2017b) proposed, tested, and compared multi-objective and single-objective optimization procedures to optimally locate sensors in sewer systems. Yazdi (2018) applied the Differential Evolution algorithm to find the best monitoring sites based on the entropy method and Banik et al. (2017a) solved the optimization problem using greedy algorithms and compared the performances with a genetic algorithm. More recently, Bayesian approaches were introduced to optimally position water quality sensors, achieving a maximum of 84% accuracy in sensor location identification (Sambito et al., 2020). However, this strategy necessitates the deployment of numerous sensors to obtain a high probability of event detection. Simultaneously, data-driven machine learning techniques, represented by clustering analysis have attracted extensive attention considering the enormous amount of data acquired by online monitoring (Zhao et al., 2016). Clustering analysis is an unsupervised learning technique that aims to partition data sets based on the similarity of samples (Jain et al., 1999). Guo et al. (2018) used fuzzy clustering analysis to automatically set monitoring points based on drainage junctions' simulated time-series water level data. Li (2021) combined agglomerative clustering and analysis of variance (ANOVA) to optimize the sensor placement under uncertain rainfalls. Existing sensor deployment strategies based on clustering analysis are mostly designed to detect flood-prone spots in combined sewer systems. Additionally, with the ultimate goal of monitoring the direct flow to the river during extreme rainfall events, Cosco et al. (2018) offered a methodology based on the normalized criterion presented to assign a certain priority to each potential monitoring point of the network.

Most of the aforementioned sensor placement studies aim to prevent and identify specific events that have an immediate impact on public life, such as illicit intrusions or urban flooding. Their inquiries focus exclusively on the hydraulic or quality information of the sewer systems. However, few studies have been conducted to optimize sensor placement for improving the routine monitoring of UDS, which requires comprehensive consideration of network hydraulic and quality performance surveillance. Whereas routine online monitoring is critical to identify and diagnose abnormal operating conditions in the network that are relatively insidious and whose effects tend to accumulate over time. Furthermore, the current hotter studies employing clustering methods recognize nodes that are representative in specific attributes (typically water level time series) to place sensors, however, practical deployment may require consideration of other criteria reflecting monitoring necessity (e.g., node topology criticality, amount of received effluent, etc.). Moreover, it is noteworthy that the common problem shared by all categories of research is the framework of optimizations still cannot be compared and assessed quantitatively based on specific indicators.

In the face of these shortcomings, this study aims to develop a novel methodology for optimizing sensor placement to improve the information content and observability of monitoring data while taking into account a more comprehensive range of practical placement factors, and thus monitor the routine operation of UDS more systematically. Unlike earlier clustering studies that only analyze node hydraulic properties, this work incorporates hydraulic and water quality information to generate sample attribute datasets to ensure the reliability of routine monitoring. In addition, based on the basic framework of traditional clustering methods, a re-clustering strategy integrating more deployment perspectives was proposed in this study. The strategy performs initial clustering with a limited cluster number, then calculates a re-clustering potential index for each cluster (the index reflects the decision-makers' preference, e.g., the average population serviced by the nodes of each cluster might serve as the index if the monitoring demand is greater in areas with higher residential density), and finally determines the necessity and cluster number for re-clustering according to the ratio of this index. In conclusion, to quantitatively illustrate the reasonableness of the clustering and re-clustering process, the concepts of joint entropy and total correlation from information theory are introduced to evaluate the obtained sensor placement solutions.

The methodology was applied to two networks with different drainage systems and sizes: the literature network Example 8 of the Storm Water Management Model (SWMM) application manual (Gironás et al., 2009), where different rainfall conditions were considered in this combined sewer system, and the real test case represented by the sewerage network of WZ city (China), which was analyzed in dry weather conditions. The paper is organized as follows: First, the sensor placement methodology is presented. Second, the test cases are introduced to assess the effectiveness of the methodology, and the results are presented. Last, the main conclusions are pointed out.

2. Methodology

The workflow of the proposed methodology is shown in Fig. 1. First, it is necessary to construct a simulation model of the study area. The water hydraulic and quality characteristics under different rainfall conditions at all nodes are obtained through the simulation of the model. Then, the initial clustering is performed on the normalized data and the re-clustering potential index is calculated for each cluster. Next, which clusters require re-clustering and the cluster number of re-clustering is determined by the proportional magnitude of this index. Finally, the two clustering results are amalgamated for the ultimate monitoring network. To assess the information enrichment of the obtained monitoring network, the time series of water level and pollutant concentration from the sensors it contains are extracted to calculate the joint entropy and total correlation. In the following sections, details of these steps and the applied algorithms, simulations, and analysis are presented.

2.1. Datasets generation and clustering model implementation

The application of a multi-objective optimization algorithm for sensor placement yields a series of Pareto optimal solutions that usually vary substantially, making the deployment adjustment implementation more challenging. In contrast, the clustering process can not only acquire the deployment scheme, but also the intrinsic connections of the nodes, which is more conducive to making timely and corresponding adjustments for unpredictable and uncertain changes in UDS. The clustering techniques applied in the sensor deployment optimization are mostly prototype-based clustering and hierarchical clustering (Guo et al., 2018; Li, 2021). Since hierarchical clustering cannot directly find the cluster center representative points, this study applied the K-means Clustering (KC), a representative algorithm in prototype clustering (Jain et al., 1999; Xu and Tian, 2015). The rules governing which samples (nodes) are merged are based on the dataset similarity, shown in the Pseudo-code of Fig. 2. In this research, the inputs for the KC model contained the water level and pollutant concentration data of all nodes generated by the Storm Water Management Model (SWMM) (Rossman, 2009) and extracted employing the "MatSWMM" toolkit (Riano-Briceno et al., 2016). The model considered a conventional pollutant COD at a fixed concentration in the dry weather flow (DWF) inflow with a fixed degradation coefficient. The average concentrations of nodes during the simulation time characterize their water quality. As for the hydraulic characteristics, the nodal maximum and average water levels as well as overflow volumes (if there is an overflow) under different rainfall conditions were considered for the combined sewer system, whereas for the separated sewage network, only the nodal maximum and average water levels under dry weather were taken into account. To simultaneously account for the influence of geographic elements, the node elevation and coordinate were additionally integrated into the attribute database.

2.2. Re-clustering potential index

The KC algorithm implements the function of dividing the nodes based on their similarity, with the most representative centroid in each cluster serving as the ideal monitoring point. However, in addition to hydraulic and quality characteristics, several practical considerations

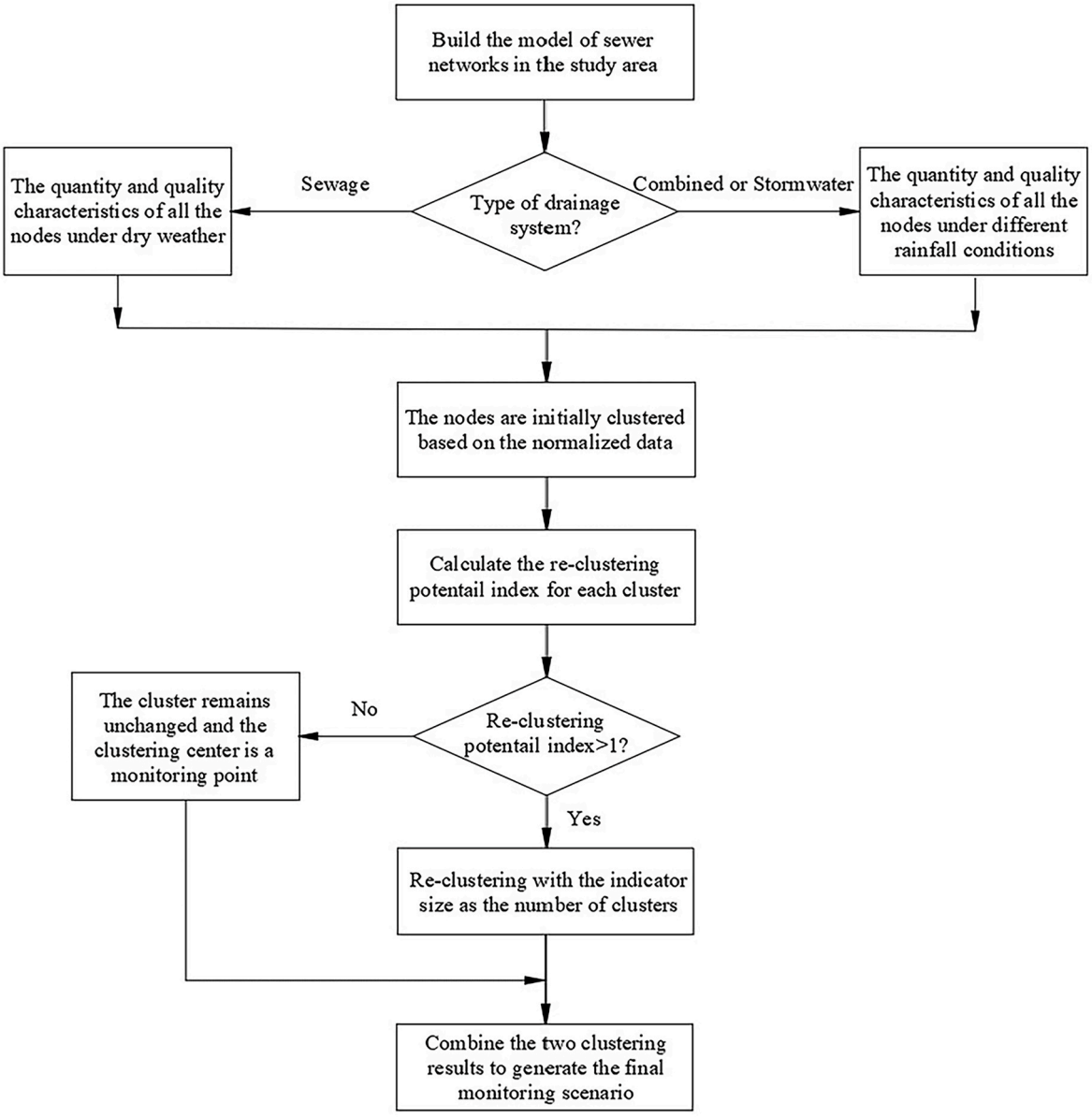


Fig. 1. The workflow of the proposed methodology.

such as potential risk and urgent priorities will be taken into account when sensors are deployed. Thus, the number of monitoring stations required for each cluster may differ, which is why the re-clustering potential index is proposed in this study. The index indicates the decision-makers' subjective preference regarding the necessity of monitoring and can be designed according to various monitoring demands. Two indices were selected as examples for specific illustrations in this paper.

ACC:

The first index Average Closeness Centrality (ACC) considers that nodes at critical topology locations should be monitored more urgently. The concept of closeness centrality (Zhang et al., 2018) is introduced here to measure a node's ability to influence other nodes in the network. The higher the closeness centrality of a node, the more central the location of the node is in the network, and the more significant it is in the network. The closeness centrality of a node is expressed by Eq. (1) as follows.

$$c(i) = \frac{1}{\sum_{j=1}^{N} d_{ij}} \tag{1}$$

Where $c(i)$ is the closeness centrality of node i, N is the total number of nodes in the network, d_{ij} is the shortest distance from node i to node j, which can be calculated by applying Dijkstra's algorithm. The metric is then normalized to obtain Eq. (2).

$$C(i) = \frac{Nc(i)}{\sum_{i=1}^{N} c(i)} \tag{2}$$

The ACC index calculates the average closeness centrality of nodes in each cluster, expressed by Eq. (3) as follows.

$$ACC_k = \frac{\sum_{i=1}^{nk} C(i)}{nk} \tag{3}$$

where ACC_k is the ACC index of the kth cluster, nk is the number of

Input: Given a dataset $D = \{x_1, x_2, ..., x_m\}$;

Number of clusters k.

Procedure:

1: Randomly select k samples from D as the initial mean vector $\{\mu_1, \mu_2, ..., \mu_k\}$

2: **repeat**

3: set $C_i = \varnothing (1 \le i \le k)$

4: **for** $j = 1, 2, ..., m$ **do**

5: calculate the distance of sample x_j from each mean vector $\mu_i \{1 \le i \le k\} : d_{ij} = \| x_j - \mu_i \|_2$;

6: determine the cluster label of x_j based on the closest mean vector: $\lambda_j = \arg\min_{i \in \{1,2,...,k\}} d_{ji}$;

7: assign x_j to the corresponding cluster: $C_{\lambda_j} = C_{\lambda_j} \cup \{x_j\}$;

8: **end for**

9: **for** $i = 1, 2, ..., k$ **do**

10: update the mean vector: $\mu_i^{'} = \frac{1}{|C_i|} \sum_{x \in C_i} x$

11: **if** $\mu_i^{'} \ne \mu_i$ **then**

12: update the current mean vector μ_i to $\mu_i^{'}$

13: **else**

14: keep the current mean vector unchanged

15: **end if**

16: **end for**

17: **until** none of the current mean vectors are updated

Output: Cluster division $C = \{C_1, C_2, ..., C_k\}$

Fig. 2. The Pseudo-code for the K-means Clustering algorithm.

nodes belonging to the kth cluster.

ANI:

The second index Average Node Inflow (ANI) considers that nodes receiving more DWF inflow may be located in areas serving higher residential densities, necessitating the addition of more monitoring stations, which can be expressed by Eq. (4) as follows.

$$ANI_k = \frac{\sum_{i=1}^{nk} NI_i}{nk} \tag{4}$$

Where ANI_k is the ANI index of the kth cluster, nk is the number of nodes belonging to the kth cluster, NI_i is the base inflow of node i.

The ratio of the index value to the minimum value can be calculated after each cluster's re-clustering potential index is obtained, and the result is rounded to the nearest integer to estimate the cluster number for re-clustering. However, for the values of node inflow with large magnitude variance utilized in the ANI index, it is more appropriate to adopt the ratio to the average value of all indices as the re-clustering number. And the two clustering results should be combined to generate the final sensor deployment arrangement.

2.3. Evaluation of monitoring network

Clustering analysis can identify the most representative nodes of hydraulic and quality characteristics with a limited number of sensors. The correlation of the same cluster efficiently eliminates economic waste, and the difference between various clusters ensures access to a broad range of network information. The joint entropy and total corre-

lation metrics based on information theory were applied in this study to verify this inference. The joint entropy (Shannon, 1948) denotes the pure information content generated by several variables and can be generalized as Eq. (5).

$$H(X_1, X_2, ..., X_n) = -\sum_{i=1}^{n_1}\sum_{j=1}^{n_2}\cdots\sum_{k=1}^{n_n} p(x_{1i}, x_{2j}, ..., x_{nk}) \log p(x_{1i}, x_{2j}, ..., x_{nk}) \tag{5}$$

The main difficulty in using the above equation to get a criterion of information content is calculating the joint probabilities $p(x_{1i},x_{2j},...,x_{nk})$. To solve this problem, the concept of total correlation (Watanabe, 1960) can be used considering dependencies among the set of variables (Eq. (6)):

$$C(X_1, X_2, ..., X_n) = \left(\sum_{i=1}^{n} H(X_i)\right) - H(X_1, X_2, ..., X_n) \tag{6}$$

To calculate $H(X_1, X_2, ..., X_n)$ using the above equation, first the entropy characterizing the uncertainty of a single random variable can be calculated by Eq. (7) as follow.

$$H(X) = -\sum_{i=1}^{n} p(x_i) \log p(x_i) \tag{7}$$

Then $C(X_1, X_2, ..., X_n)$ can be computed by estimating mutual information from k-nearest neighbor statistics (Kraskov et al., 2003). In this study, the variable X_i denotes the water level or water quality time series of the ith monitoring point in a deployment scheme. The time series is transformed into "pulses" of discrete information through the quantization process. It rounds a value x to its nearest lowest integer multiple of a namely x_q, which can be expressed as Eq. (8) (Alfonso, 2010).

$$x_q = a \cdot floor\left(\frac{2x + a}{2a}\right) \tag{8}$$

From the information point of view, an optimal monitoring network should satisfy at least two basic objectives: maximum information content at all gauging sites and minimum dependency between them. Correspondingly, a monitoring network with greater joint entropy and smaller total correlation can be considered ideal. However, the re-clustering process, which takes into account additional deployment concerns, is not entirely dependent on the hydraulic and quality similarity of nodes, and therefore may result in some information losses. The details will be given in the case studies section that follows.

3. Case studies

In order to compare the procedures, two networks with different drainage systems and sizes were used as case studies. One is a small example of the SWMM manual and the other one is a real network in China. Details of both networks are provided below. According to research in the literature, the COD concentration of the DWF inflow in the presented examples was set at 300 mg/L with a degradation coefficient of 0.2 in the models (Guohui et al., 2011). And elevations, coordinates, hydraulic and water quality data were all included in the node's attribute dataset.

3.1. Example 8 network

The small network in literature Example 8, presented in the SWMM reference manual, is a combined sewer system that conveys both sanitary sewerage and stormwater through the same pipes (Fig. 3). The

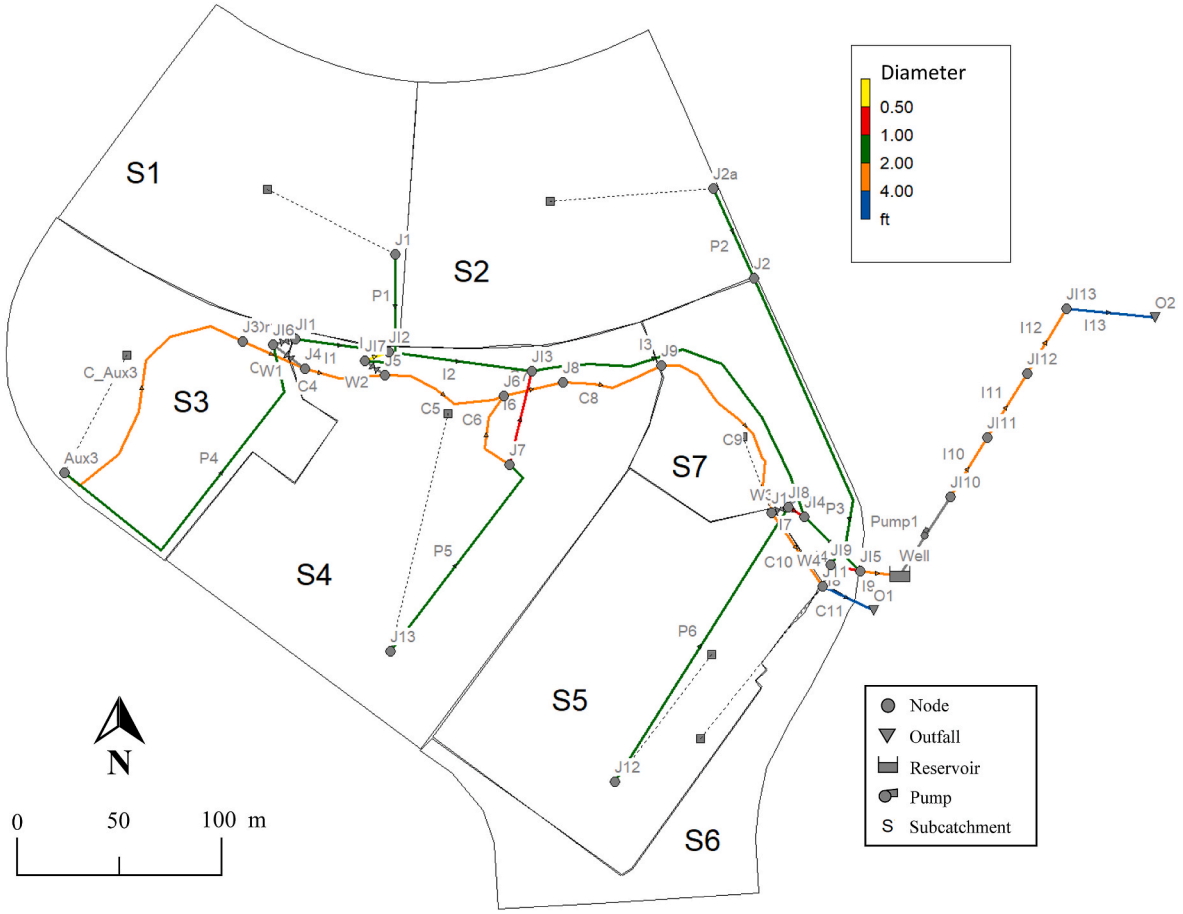


Fig. 3. Example 8 network scheme.

network serves an area of 0.12 km^2 and consists of 31 nodes (28 junctions, 2 outlets, and 1 storage unit) and 35 links (29 conduits, 1 pump, 1 orifice, and 4 weirs). The two outlets are the wastewater treatment plant (O2) and the overflow (O1) during wet weather. All geometric data and the dry weather inflows, expressed as daily mean values, have been reported by Gironás et al. (2009). The nodal maximum and average water levels and overflow volumes (if there is an overflow) under different rainfall conditions (2-year, 10-year, and 100-year rainfall gauges) were considered in this case. The implementation of the re-clustering strategy was not taken into account because the goal of this case study is to validate the compatibility between the clustering process and the introduced informatics evaluation metrics, which provides a supportive basis for information theory-based evaluation techniques in this study.

3.2. X network

WZ is a city located in the southeast coastal district of China with a subtropical monsoon climate. X is a small industrial park in WZ, which has a separate sewerage network shown in Fig. 4. The study area covered 2.63 km^2 with a mean relative elevation of 5.06m. The area is relatively flat with an elevation gap of less than 3 m. The system is divided into 785 sub-catchments and served a population of 36,852 (2020) with an approximate amount of daily produced wastewater of 7370.4 m^3. The scheme consists of 785 circular conduits connecting 786 nodes (785 junctions and 1 outfall). The daily mean values of the dry weather flows (DWF) in the nodes were assigned considering the population connected to each node. All geometric data, not reported herein, can be requested from the authors. Only the nodal maximum and average water levels under dry weather were considered for the hydraulic characteristic. The other parameters matched the case from example 8 exactly. Following the confirmation of the rationality of the informatics metrics assessment system in Case 1, this case utilizes it to investigate the effectiveness of the re-clustering strategy, and thus the two re-clustering indices proposed in the previous section were examined here.

4. Results and discussion

The general performances of the methodology without the re-clustering process were reported first for the small network and further the re-clustering extension was applied to the real network, where more comprehensive experiments were run and reported.

4.1. Example 8 Network

Only the nodes of underground pipes in the combined sewer system were considered potential monitoring points (19 in total). First, we investigated the clustering results generated with different numbers of sensors. Taking into account the limitation of the actual number of sensor locations, the scenarios of 2–5 sensors were selected for the initial experiment in this case, with the monitoring site arrangement shown in Fig. 5 and the quantitative evaluation shown in Fig. 6. Only the informatics indicators of water level time series were calculated here since there are only a few nodes with DWF inflow in this case, resulting in a lack of reference significance for the information entropy evaluation of water quality data. The value of the parameter a can be seen as the minimum dimensional unit of water level, which in this case was set at 0.01. As shown in Fig. 6, the comparatively most appropriate number of monitoring sensors is 3, which has the highest joint entropy H value, and the total correlation C value is kept within an acceptable range, preventing economic waste. What might be inconsistent with theoretical expectations is that the H value does not always increase with the number of sensors. This is because the computation of H is based on the C values. Extremely large node dependencies, like in the case of 4 and 5 sensors, will have a detrimental influence on the H value. This phenomenon can serve as a theoretical basis for determining the number of monitoring stations. In addition, observation of the different schemes under different rainfall conditions revealed no uniform pattern, therefore we extended the investigation of this phenomenon by adding rain gauge situations in the subsequent comparison of different schemes with 3 sensors.

According to the clustering results, the three sensors are arranged at nodes J1, J7, and JI12. Specific analysis of the result reveals that the nodes in the cluster represented by J1 (J1, J2a, J12, J13) are all runoff

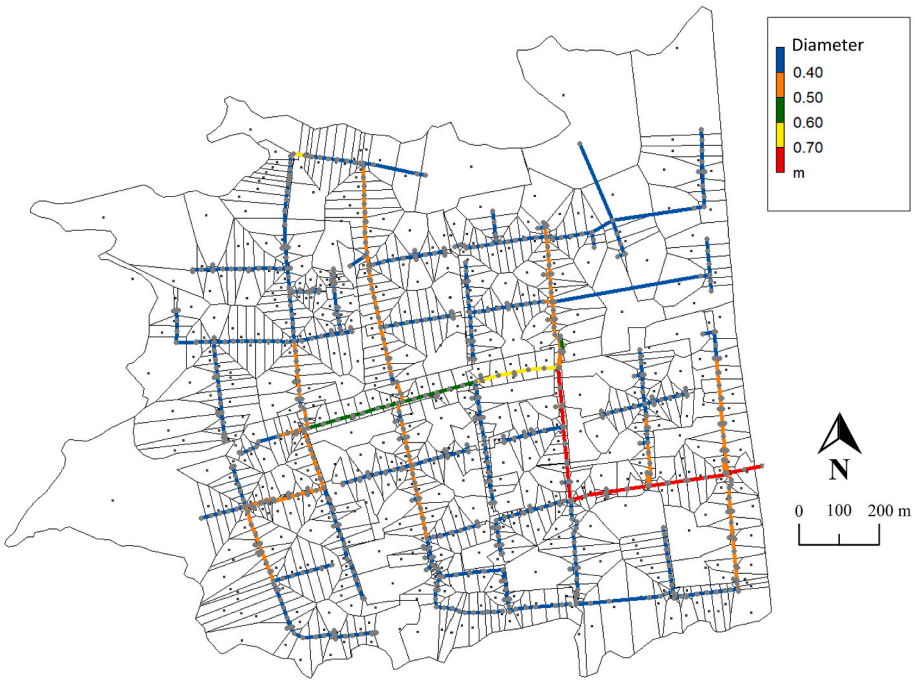


Fig. 4. X network scheme.

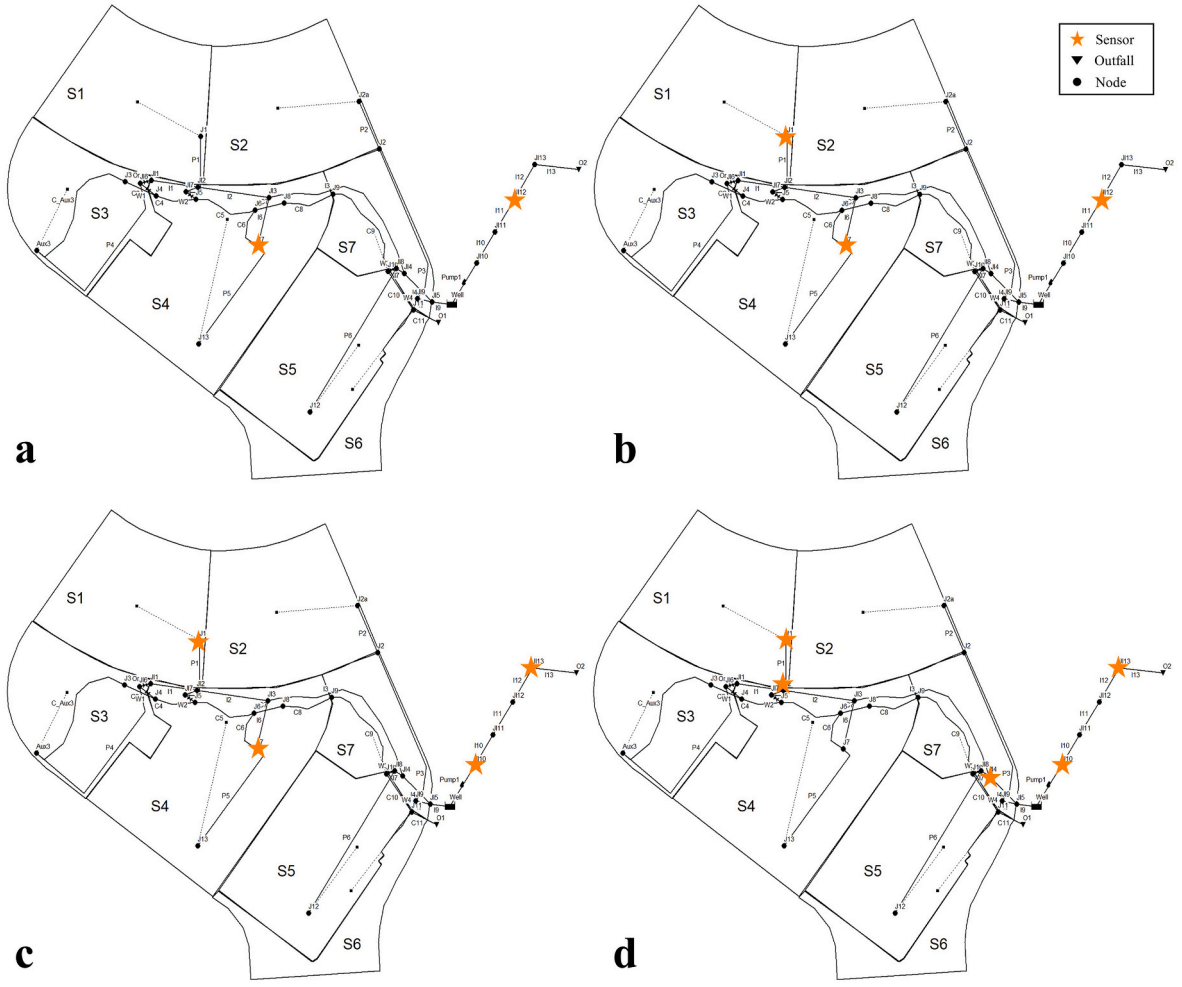


Fig. 5. Optimal layout of monitors for different sensor numbers (Example 8 network): (a)two sensors; (b)three sensors; (c)four sensors; (d)five sensors.

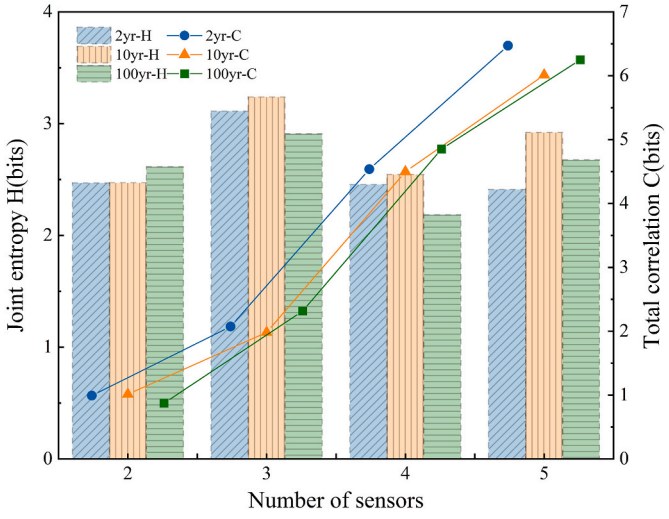


Fig. 6. Quantitative evaluation of optimal deployment schemes for different sensor numbers (Example 8 network).

inlet points, the cluster represented by JI12 (JI10, JI11, JI12, JI13) contains all pressure pipe nodes, while the other common nodes form a cluster with the representative point JI1 (J2, J7, JI1, JI2, JI3, JI4, JI5, JI6, JI7, JI8, JI9), which presents an obvious difference in nodes' characteristics. To quantitatively assess the contribution of clustering methods in expanding the information content of the monitoring network, we compared the joint entropy and total correlation of the clustering result with any other three-point placement schemes (Table 1).

In this study, two schemes were selected for each of the three monitoring points from three clusters, two clusters and one cluster (a total of six) and compared with the scheme generated through clustering. The 5-year and 50-year rainfall gauges were introduced to explore the impact of weather conditions more comprehensively. The data visualization of Table 1 is displayed in Fig. 7 as can be seen, the results obtained by the clustering process have a clear advantage over the other schemes (higher H and lower C). And the schemes of the deployment distribution belonging to three clusters are overall better than those belonging to two clusters, while those belonging to the same cluster perform the worst due to their high dependency. These phenomena demonstrate the superiority of the clustering approach in terms of increasing the information coverage of the monitoring network. Further investigation of the superiority and inferiority of each scheme under different rainfall conditions finds no consistent rule, indicating that the capacity of different deployment schemes to capture information under diverse weather scenarios varies. This information can be the theoretical basis for selecting monitoring solutions based on the local climatological conditions.

Since this case was discussed in the previous studies related to optimal sensor placement, the results were also analyzed in comparison with the previous studies in this paper. The published studies discussing this case were all aimed at water quality monitoring, and two

Table 1
Quantitative evaluation of different deployment schemes for three monitoring sensors (Example 8 network).

Rainfall conditions	H or C (bits)	Clustering result	In three clusters		In two clusters		In one cluster	
		(J1, J7, JI12)	3—① (J2, JI12, J2a)	3—② (JI4, JI10, J13)	2—① (JI1, JI5, JI10)	2—② (JI5, JI1, J12)	1—① (JI1, JI3, JI5)	1—② (JI6, JI7, JI8)
2-yr	H	3.112	2.979	2.654	2.423	1.092	0.586	0.349
	C	2.072	2.294	2.591	2.858	4.169	4.602	4.474
5-yr	H	3.077	2.892	2.674	2.530	1.177	0.820	0.487
	C	2.133	2.402	2.585	2.776	4.083	4.378	4.338
10-yr	H	3.238	2.973	2.884	2.669	1.610	0.980	0.536
	C	1.981	2.325	2.390	2.648	3.662	4.242	4.320
50-yr	H	3.235	2.830	2.971	2.623	0.923	0.351	0.606
	C	1.996	2.501	2.315	2.688	4.343	4.856	4.254
100-yr	H	2.910	2.850	2.748	2.853	0.875	0.531	0.591
	C	2.317	2.462	2.524	2.444	4.393	4.670	4.293

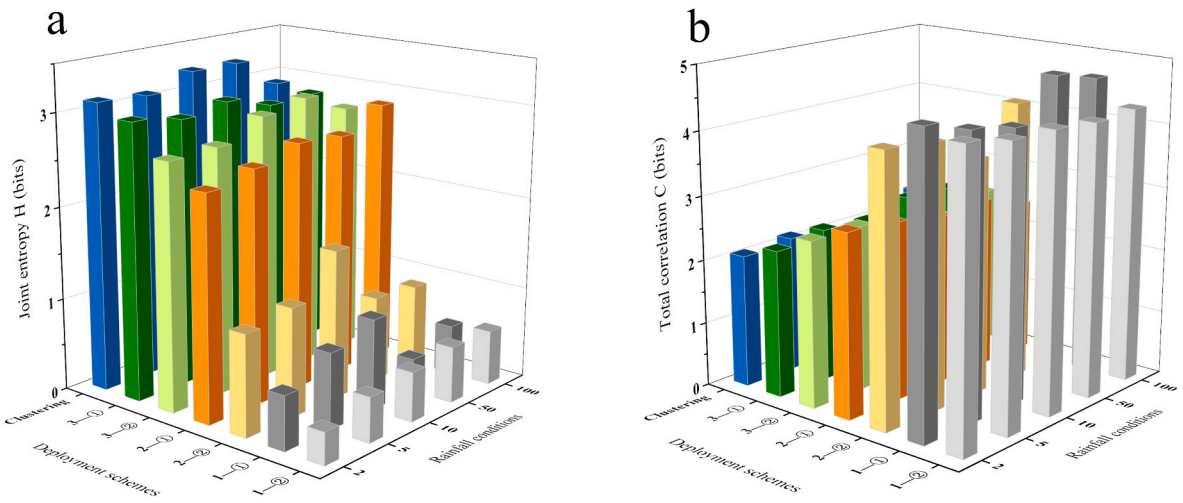


Fig. 7. The visual presentation of the results in Table 1: (a)Joint entropy H; (b) Total correlation C.

representative studies applying the genetic algorithm (NSGA-II) for multi-objective optimization (Banik et al., 2017b) and Bayesian decision network (Sambito et al., 2020) were selected to compare with the clustering strategy of this research. The arrangements of three sensors obtained by the three different methods are shown in Fig. 8. It can be seen that the monitoring stations determined by the genetic algorithm and Bayesian method are concentrated in the downstream region of the drainage system, while the placement scheme generated in this paper is more decentralized and can simultaneously acquire information from the upstream, interceptor and downstream regions, providing a wider monitoring coverage and a better problem detection capability.

4.2. X network

All junctions in the X sewage network were considered potential monitoring points (785 in total). Re-clustering methods were applied and evaluated in this case. The initial number of clusters in the experiment was set to a sequence from 8 to 30 with an interval of 2, and the two re-clustering potential indices ACC and ANI were both discussed in this case. Since it is a separate loop sewerage network with DWF inflow at practically all nodes, both water level and water quality data were analyzed informatically. For computing the information entropy of water levels, the parameter a is set to 0.001, while it is set to 0.005 when processing the data set of pollutant concentration.

The results of applying ACC as the re-clustering potential index are shown in Table 2. For each initial cluster number, this study calculated the values of informatics metrics for the results after re-clustering and the direct clustering results consistent with the total cluster number after the re-clustering procedure. For instance, the re-clustering method increased the initial cluster number of 8–9 monitoring points, therefore the scheme produced by clustering directly with 9 as the cluster number was also assessed for comparison. Firstly, a vertical comparison of direct clustering schemes with different sensor numbers reveals that, due to the large scale of the network in this case, within a relatively limited number of monitoring stations, there is a general trend that the more clusters there are, the greater the H-value and the C-value increase is limited, i.e., the more sensors there are, the more capable the monitoring network is of acquiring information. Further internal comparison of the re-clustering results with the direct clustering results reveals that the information content of the water level and pollutant concentration parameters do not necessarily behave in the same pattern, hence the total of H values and the sum of C values were computed separately for the two sequences (Fig. 9). As observed in Fig. 9, the informatics performance of the re-clustering scheme has an advantage over the direct clustering scheme in most scenarios, except for the initial cluster numbers of 16 and 24, where the higher C values result in a slightly inferior overall informatics performance than the direct clustering scheme. As previously stated, the principal difference between the re-clustering procedure that integrates other deployment considerations and the pure clustering strategy may result in certain information losses. Since anomalies in pipes close to the layout center have a wider impact on the system and are more challenging to repair, it is necessary to emphasize monitoring station data located near the center. This is why the ACC index was introduced. Taking the example of sensor deployment layouts with 16 initial clusters for a relatively larger gap in informatics metrics (Fig. 10), the result after re-clustering does contain

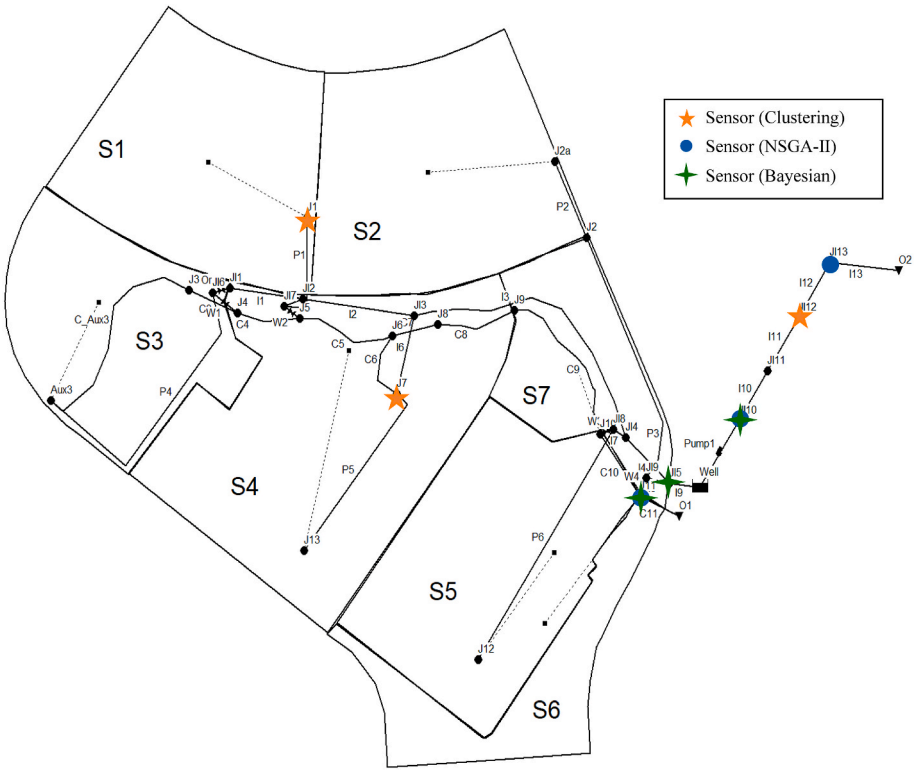


Fig. 8. The deployment scheme of 3 sensors generated by different methods.

Table 2
Quantitative evaluation of different deployment schemes applying the ACC index.

Initial clustering number (final number)	Water level (bits)				Water quality (bits)			
	Re-clustering		Direct clustering		Re-clustering		Direct clustering	
	H	C	H	C	H	C	H	C
8 (9)	10.678	4.365	11.105	3.625	12.045	0.703	9.656	0.926
10 (11)	12.785	3.363	11.908	4.466	11.919	0.207	11.529	0.548
12 (14)	17.442	5.593	16.152	5.203	16.963	1.114	15.274	2.046
14 (16)	18.181	6.523	18.688	5.791	18.467	2.165	17.638	1.164
16 (19)	22.262	7.034	26.325	3.172	20.434	1.730	24.499	1.013
18 (23)	28.679	7.066	26.595	8.561	27.158	1.365	28.447	2.596
20 (24)	33.514	3.424	27.806	9.182	30.229	1.724	26.803	4.279
22 (29)	39.398	6.072	33.078	12.178	36.123	1.535	37.901	3.440
24 (32)	37.522	12.690	39.799	10.450	39.443	4.888	39.341	6.038
26 (34)	44.938	8.662	43.479	10.447	44.996	4.676	42.841	5.534
28 (38)	54.280	6.220	50.886	6.993	51.268	1.332	50.531	4.473
30 (40)	56.081	6.703	48.474	14.360	47.122	4.277	52.699	3.136

more nodes with higher Closeness Centrality (CC), which intuitively fulfills our intention. In summary, the most desirable solution has an initial cluster of 28 and subsequently 38 monitoring points (highest H-value with a low C-value), and if the budget is limited, a solution with an initial cluster of 20 or 22 may be preferred (significant advantages over direct clustering).

The experimental outcomes of applying ANI as the re-clustering potential index are shown in Table 3. Table 3 displays approximately similar information to Table 2 of applying the ACC index, still presenting a significant trend that the performance of the informatics metrics improves with an increasing number of sensors, and the informatics metrics trends of water level and pollutant concentration series still show irrelevance. Thus likewise, the sum of the metrics for the two sequences was calculated (Fig. 11). However, the informatics performance of the re-clustering scheme applying the ANI index is worse and inferior to the direct clustering scheme in more cases compared to ACC, demonstrating that different re-clustering potential indices affect the initial clustering process to varying degrees. The ANI index was introduced with the consideration that the nodes receiving more DWF serve more inhabitants and their monitoring must be enhanced to ensure the efficient operation of the wastewater system. Similarly, when the sensor deployment layouts of the 18 initial clusters with relatively larger disparities from the direct clustering method were examined (Fig. 12), the re-clustering strategy generated a deployment scheme that was overall closer to the nodes having higher DWF, which is consistent with the initial intention of introducing ANI. To conclude the scenario of applying the ANI index, the increase in H-value becomes limited and slow when the initial cluster number reaches 20, hence the final scheme of 28 sensors is recommended to balance the informativeness and economics.

The informatics performance of different re-clustering potential indices varies with different initial sensor numbers, and the assessment of deployment schemes from diverse perspectives may also lead to different conclusions, as indicated by the above two indices, which serve

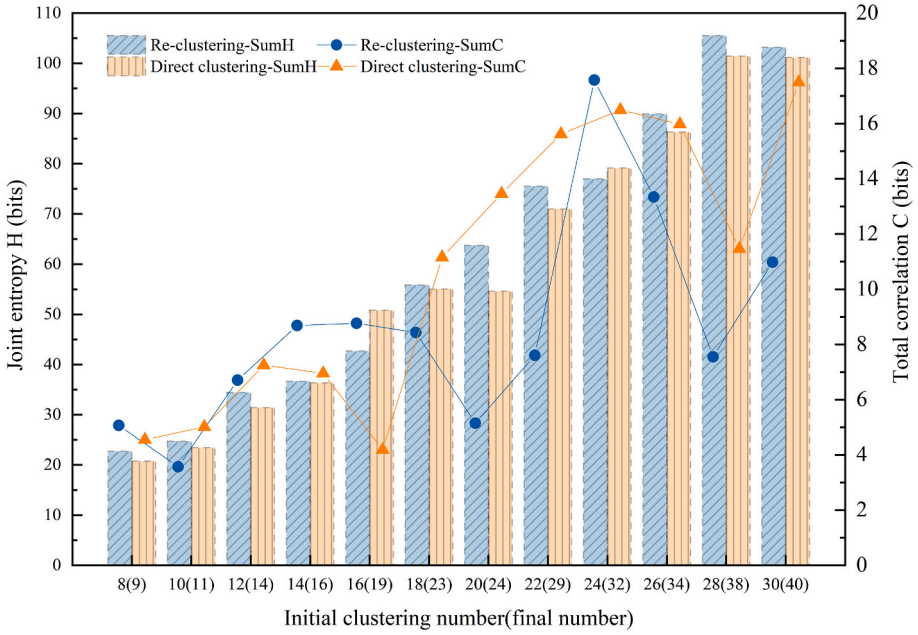


Fig. 9. Quantitative evaluation of deployment schemes combing water level and quality results (ACC index).

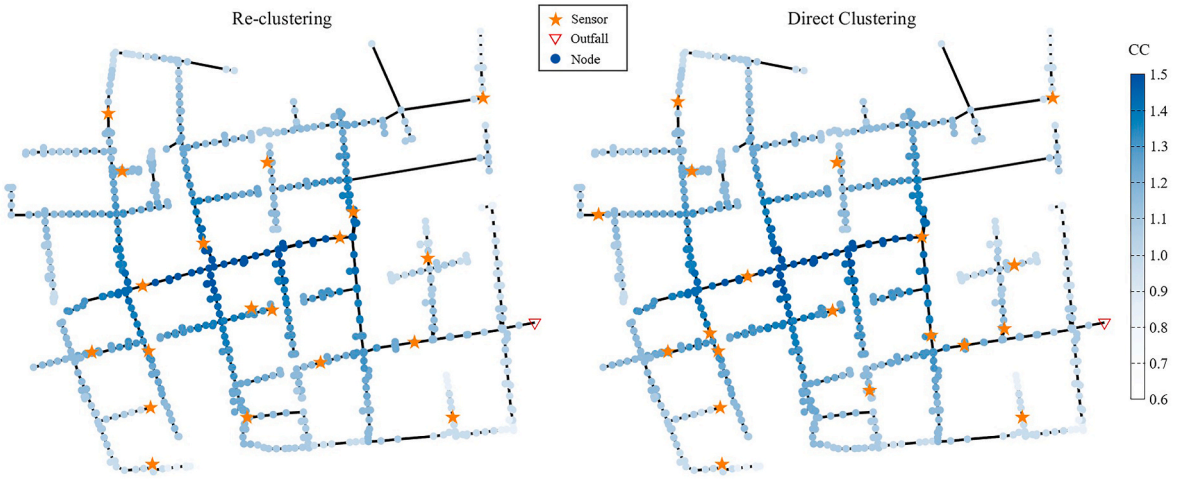


Fig. 10. Sensor deployment layouts with the initial clustering number 16 (ACC index).

Table 3
Quantitative evaluation of different deployment schemes applying the ANI index.

Initial clustering number (final number)	Water level (bits)				Water quality (bits)			
	Re-clustering		Direct clustering		Re-clustering		Direct clustering	
	H	C	H	C	H	C	H	C
8 (10)	12.847	3.752	12.048	2.418	12.459	0.537	10.309	0.135
10 (13)	16.111	3.136	14.941	4.845	18.154	0.223	14.276	1.336
12 (16)	20.320	5.689	18.688	5.791	19.039	0.657	17.638	1.164
14 (18)	21.339	6.351	20.942	6.673	20.246	2.055	21.378	0.884
16 (21)	25.355	6.526	24.895	7.714	20.971	1.168	25.198	2.058
18 (25)	29.865	8.026	35.508	3.387	27.887	0.937	32.535	2.178
20 (28)	39.819	3.016	34.764	8.823	37.175	1.059	33.456	3.532
22 (30)	40.954	5.469	41.761	5.126	35.023	1.003	37.593	9.298
24 (32)	38.221	11.276	39.799	10.450	38.362	4.288	39.341	6.038
26 (34)	42.414	10.320	43.479	10.447	40.529	5.463	42.841	5.534
28 (37)	46.166	10.922	45.348	10.919	44.643	3.532	44.007	3.173
30 (40)	56.784	5.436	48.474	14.360	49.120	3.818	52.699	3.136

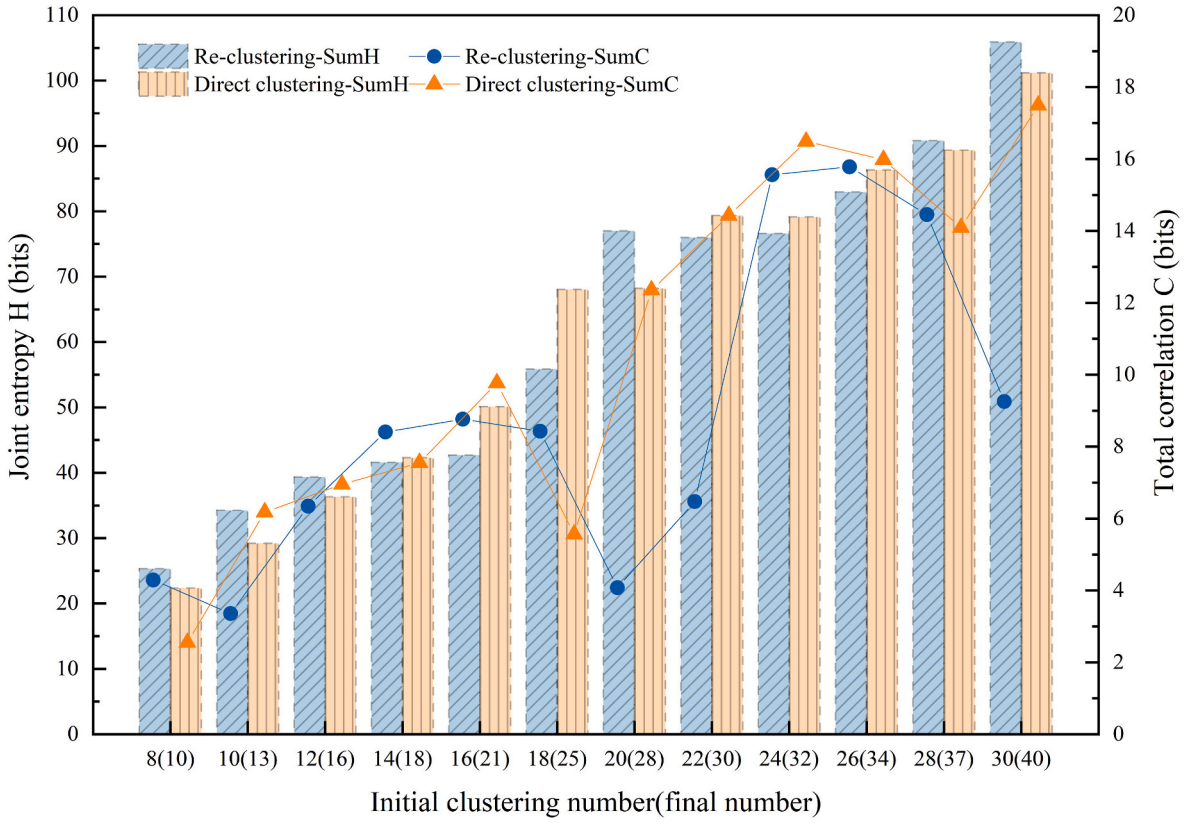


Fig. 11. Quantitative evaluation of deployment schemes combing water level and quality results (ANI index).



Fig. 12. Sensor deployment layouts with the initial clustering number 18 (ANI index).

as illustrations of the re-clustering methodology framework. However, in general, the re-clustering strategy proposed in this research is more comprehensive and practical since it can take into account the influence of more relevant factors for the actual deployment while guaranteeing the information richness of the monitoring network.

5. Conclusions

This study developed a methodology for optimizing sensor placement to monitor the daily operation of drainage networks. The hydraulic and water quality attributes of the nodes were incorporated into the basic clustering framework to overcome the limitations of the standpoint on optimal sensor placement that has been considered in the latest research. Also, in view of the fact that the clustering process selects locations just from the similarity of nodes' attributes, this study proposed a re-clustering strategy to integrate the effects of other practical factors such as node topological significance and service scale. The strategy applies the K-means algorithm for the initial clustering, then calculates the specified re-clustering potential indices (ACC and ANI) of each cluster for re-clustering, and lastly merges the two clustering results to derive the final deployment scheme. To quantify the rationality of the obtained scheme, the joint entropy and total correlation metrics based on information theory, which reflect the information content, were introduced. The methodology was applied to two networks with different drainage systems and sizes: the combined sewer system Example 8 in the literature and the real separated sewerage system X

network (China). The results of the Example 8 case show that the 3-sensor arrangement is the most reasonable and the clustering result outperforms other placements that violate the clustering principle, demonstrating that the clustering process succeeds in its objective of expanding the information contained in the monitoring network. And the results for the X network present an overall pattern that the informatics performance of the monitoring network improves with the number of available sensors. The application of the ACC and ANI indices yields a slightly different scheme performance, while the ultimate scheme with an initial clustering number 20 for both indices is the more preferable choice. Additionally, it can be inferred from the comparing results with the direct clustering method that, in general, the re-clustering strategy is a more comprehensive and operational approach that encompasses more practical concerns while guaranteeing the information richness in the monitoring network.

It is worth mentioning that both re-clustering potential indices applied in this paper are relatively straightforward and might compromise the effectiveness of the derived deployment schemes' informatics performance. Therefore, it is recommended to investigate more precise and rigorous mathematical models or a more rational way of determining the re-clustering number to make the process more reliable.

Credit author statement

Siyi Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – original draft, Writing – review & editing. **Xiangwei Zhang**: Data curation, Resources, Funding acquisition. **Jiaying Wang**: Data curation, Funding acquisition. **Tao Tao**: Resources, Data curation, Supervision, Project administration, Funding acquisition. **Kunlun Xin**: Resources, Supervision. **Hexiang Yan**: Resources, Supervision. **Shuping Li**: Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was financially supported by the National Natural Science Foundation of China (NSFC) [Grant 51978493] and [Grant 51778452].

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